

ATRIUM TNA Report on “Computer Vision in Archaeology” Training School in Brno, Czech Republic

By Aaron Grados

The summer school at Brno about CV in archaeology was one of the best experiences in my professional career so far. Despite being fast-paced the topics covered were enough to give us an introduction to computer vision and its applications in archaeology, but it was more than that, because the people giving us the lectures were also experienced in these topics, so talking to them about our projects helped a lot to look for ways to improve our approaches.

From 14th to 18th of September we through an intensive training in computer vision, where we learned the basics about python and coding with agents and the applications of computer vision in artifacts images, microscopy images, satellite images, rock art images and coins images and 3D models.

Day 1

We started the week with an introduction to everything computer vision in archaeology and the basics of python to understand the code we can generate with coding agents.

The first block (introduction) was appreciated because it gave us all an starting point to follow the next topics despite our backgrounds, but the second topic was a little more difficult to digest. I could follow because I have some background in python, but if you were not familiar with it, it may have been very overwhelming as it was too much information with little time to process it. As a suggestion for future opportunities, maybe provide the students with some extra material about python in advance so it can be read or watched before the class and the topic becomes a little more understandable. I know the idea is not to be proficient in python but the language is really important in CV, and despite the fact that a coding agent can do it now, it helps to understand a little what the agent is writing.

The last section about coding with agents was also a good starting point for the week since it helped a lot with writing and debugging what we

were going to be doing the rest of the week in our Jupyter notebooks and collab sessions.

Day 2

Artifacts datasets and annotation. The perfect way to start in my opinion, because image annotation is basic for the work we will be doing, and they showed us a great method to make the work easier, using SAM2 with cvat. I had never annotated an image, so this was an essential topic for me, and I got to try it in my own dataset, where it worked out great.

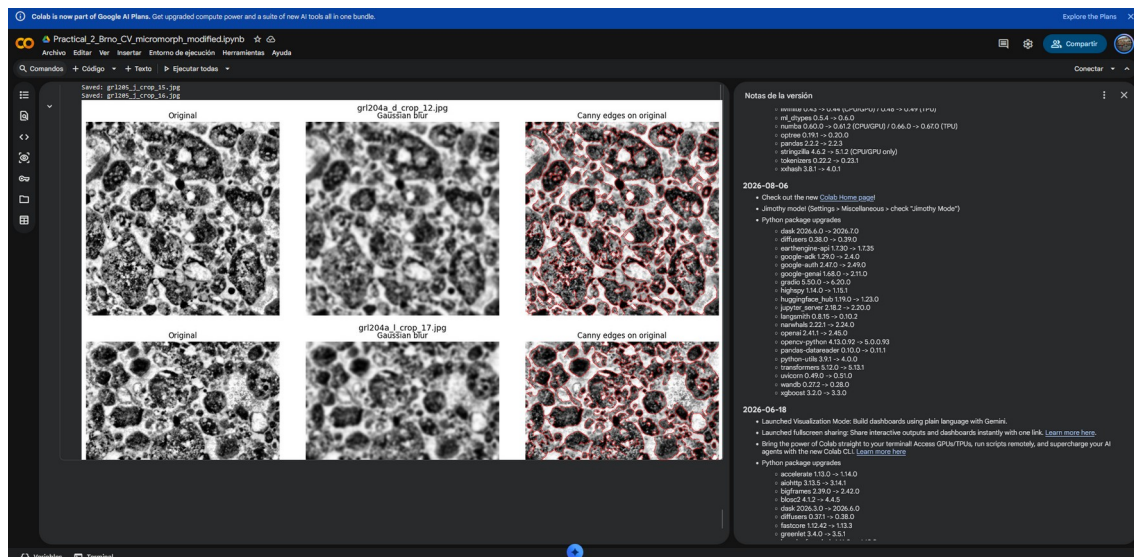
Now, artifacts are the most common source of work for archaeologist, so having this topic as the first one about computer vision was spot on. It allowed me to think about the work I want to do with my datasets and how to approach it better.

The second part of the day was covered by the students, and this was the perfect opportunity to see what other people were working on and learn more about different aspects of applied computer vision.

Day 3

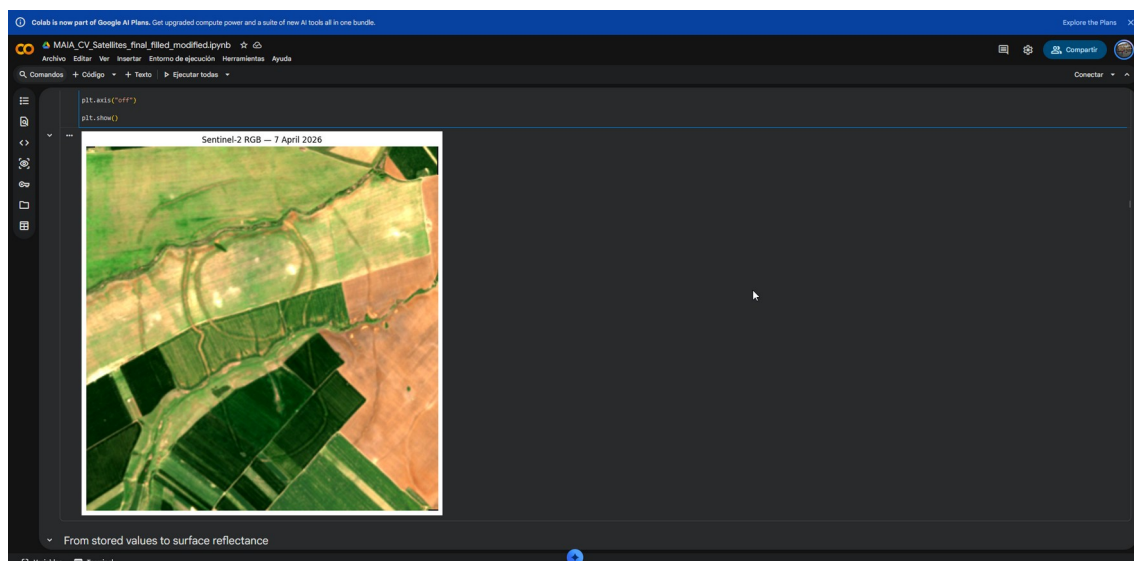
Microscopic image datasets was the topic of the whole day, and despite the fact that I've never thought about working on microscopic data, the topic was so interesting and well presented that even if you were not to do what the lecturer presented in the same type of dataset you could take the basics of it and apply it to another image dataset.

I really appreciate the practice with Jupyter notebook as it made us look for the answers in the documentation of the repos instead of just having us run the code and see what happens.



Day 4

Two topics were covered on this day as well as a lab tour. The first block was about satellite images, and the approach to recognize the features was doing image enhancement first. The tools used for this can be applied to other datasets, so learning about this and how to use them was right. But to recognize the features properly image enhancement is not enough, so we complement it with GIS data, where we used shapefiles that had geometry in the areas where a feature could be, so it helped the model recognize the features more precisely.



The second part was about rock art, a topic close to me since I've worked with rock art with photogrammetry and color manipulation. The idea of

the presentation was for the tools to help us find “invisible” designs, and we also had a Jupyter notebook practice to identify cupmarks.

The idea that I liked the most about this presentation was the use of a model like YOLO and fine-tuning it to find the features we need, as this an approach that can be replicated for other projects and I was not aware of that.

Day 5

The last day of school, but not the least important, as we saw the work on coins dataset. But what I think was brilliant about this presentation was the showcase of the thought process of a complex problem, asking us questions on how we would solve it and going step by step on the solution of the problem, mixing everything we’ve seen so far and showing us how one tool may not be the only solution but mixing tools from different areas, and using the help of Hugging Face to search for possible solutions to our problems by thinking outside the box.

Conclusion

I want to thank everyone involved in the development of the course, it was a great experience for someone interested in digital archaeology with little to none chances of getting to know other digital archaeologists in its home country. I want to thank also my newly found friends for making this week as enjoyable as possible with discussions on different topics, from professional to silly topics, it was a pleasure.